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## Syllabus

### Data Visualization Module

#### Lecture 1: Introduction to the course

Introduction to the Data Visualization module; textbooks; definitions and terminology; visual perception; pre-attentive attributes; Gestalt principles

[slides]

**reading material:** Chapter 1 (Munzner 2014)

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#### Lecture 2: Nested model

Analysis framework: nested model; data abstraction (what); common types of data; task abstraction (why)

[slides]

**reading material:** Chapter 2,3,4 (Munzner 2014)

**resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 2]; [video Chapter 3]; [video Chapter 4]

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### Lecture 3: Visual encoding

Visual encoding; marks and channels; color in visualization; color palette; color deficiency; color spaces

[slides]

**reading material:** Chapter 5: Marks and Channels, Chapter 10: Map Color and Other Channels (Munzner 2014)

**resources:** VizPalette **resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 5-a]; [video Chapter 5-b] [video Chapter 10-a]; [video Chapter 10-b]; [video Chapter 10-c]

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### Lecture 4: Common charts

Visualize tabular data; common visual idioms and charts; scatterplot; (stacked) bar chart; streamgraph; dot/line chart; Gantt chart; slopegraph; heatmap; radial bar chart; star plot; radar plot; pie chart; coxcomb chart; parallel coordinates; dual-axis charts; Visual vocabulary

[slides]

**reading material:** Chapter 7: Arrange Tables (Munzner 2014)

**resources:** Visual Vocabulary **resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 7-a]; [video Chapter 7-b]

### Lecture 5: Static plotting in Python

Static plotting in Python; basic plotting in matplotlib; style and personalize plots; towards more advanced plotting in matplotlib; introduction to seaborn; rules of thumb in data visualization

[slides-a] [slides-b]

**tutorials:** - 01-matplotlib/01-figures\_subplots.ipynb — figures and subplots - 01-matplotlib/02-plotting.ipynb — basic plotting - 01-matplotlib/03-customizing\_appearance.ipynb — style and customisation - 01-matplotlib/04-visual-encoding.ipynb — visual encoding: marks, channels, and color - 01-matplotlib/05-advanced\_subplotting.ipynb — advanced subplotting - 01-matplotlib/06-charts-matplotlib.ipynb — common charts (scatter, bar, line, histogram) with Gapminder

**resources:**

- matplotlib: Examples | Tutorial | User guide
- seaborn: Gallery | Tutorial | API

## Lecture 6: Geographical plotting in Python

The importance of spatial thinking; exploring spatial phenomena visually; how to create informative thematic cartography; main thematic cartography types and rules of thumbs; pitfalls of using spatial data in computational disciplines

[slides]

## Lecture 7: Visualize static networks in Python

Visualize static networks using NetworkX and Matplotlib

## Lecture 8: Interactive Views

How to handle complexity; manipulate: change, select, navigate; facet: juxtapose, partition, superimpose; reduce: filter, aggregate, embed

[slides]

**tutorials:** - 02-interactive/01-plotly-express.ipynb — interactive charts with Plotly Express (same 4 charts + animated bubble chart) - 02-interactive/02-altair.ipynb — Grammar of Graphics with Altair (linked selections, layering)

**resources:** Tamara Munzner, Visualization Analysis & Design class, 2021 [video Chapter 11-12a] [video Chapter 11-12b] [video Chapter 13] [video Chapter 14]

## Lecture 9: Introduction to D3.js

Why D3.js, why now: when declarative grammars run out; the web platform in 10 minutes (HTML, CSS, JavaScript, the DOM, SVG); SVG attributes as visual channels; D3.js in three ideas (selections, scales, joins); the margin convention; building the Gapminder scatterplot end-to-end; Rosetta stone — the same chart in matplotlib, Altair, and D3.js

[slides]

**tutorials:** - 09-d3js-intro/09a-building-svg-with-d3.html — HTML scaffolding, select/append/attr/style, SVG element catalog, coordinate system, attribute playground - 09-d3js-intro/09b-data-driven-svg.html — the data join (selectAll · data · join); explicit enter / update / exit decomposition - 09-d3js-intro/09c-scales-and-axes.html — scales (linear / log / band toggle), margin convention, axes; same skeleton on Gapminder - 09-d3js-intro/09d-loading-data.html — d3.csv accessor; helpers (extent, group, rollup); formatters (d3.format, d3.timeFormat) - 09-d3js-intro/09e-first-scatterplot.html — capstone: full Gapminder scatterplot end-to-end

**reading material:** - Murray, *Interactive Data Visualization for the Web* (2nd ed.), chapters 1–6 — free online - Bostock, “Thinking with Joins”

**resources:** d3js.org | Observable D3.js collection

Legacy slide decks dv09-D3-course-lesson-1.pdf and dv09-D3-course-lesson-2.pdf (third-party material) are kept under `slides/` as supplementary reference.

## Lecture 10: Advanced D3.js — Update Pattern, Transitions, Interactions, Layouts, Zoom

The five moves that turn a static D3.js chart into a living one: **change** (the update pattern with `selection.join` and key functions), **time** (transitions: duration, delay, easing, chained, staggered, shared timeline), **input** (event listeners and the event → state → render loop; hover, click-toggle, slider, brush), **composition** (layouts: bar chart with `scaleBand`, stacked bar with `d3.stack`, streamgraph with offsets and curves), **navigation** (`d3.zoom` rescale-the-scales pattern with clip path)

[slides]

**tutorials:** (index) - 10-d3js-shapes-layouts/10a-update-pattern.html — General Update Pattern; `selection.join`; key functions vs index joins; custom enter/update/exit callbacks; sortable bar chart - 10-d3js-shapes-layouts/10b-transitions.html — transitions, easing, chaining, staggered delays, enter/exit transitions, bar chart race - 10-d3js-shapes-layouts/10c-interactions-capstone.html — event → state → render loop; hover, click-toggle, year slider, brush; interactive Gapminder capstone - 10-d3js-shapes-layouts/10d-examples-gallery.html — layout-driven charts: bar chart, `scaleBand` padding, stacked bar with `d3.stack`, streamgraph; *which layout for which question?* decision table - 10-d3js-shapes-layouts/10e-zoom-and-navigation.html — `d3.zoom` behaviour; rescale-the-scales pattern; clip path; production checklist

**examples:** 24 self-contained HTML demos in `examples/` (linked from the index above), one per concept.

Legacy slide decks dv10-D3-course-lesson-3.pdf and dv10-D3-course-lesson-4.pdf (third-party material) are kept under `slides/` as supplementary reference.

## Lecture 11: Networks with D3.js — NetworkX→D3.js, Force Layouts, Interactivity, Encoding, Rendering & Scale, Web Workers, Capstone

From the NetworkX/D3.js JSON interchange to a complete interactive network viewer: **data** (`nx.node_link_data` and `d3.json`, the nodes/links contract, source/target string-then-object lifecycle), **layout** (`d3.forceSimulation` with `forceLink`, `forceManyBody`, `forceCenter`, `forceCollide`; tuning strength and

distance; pinning nodes; cluster forces), **interactivity** (drag, hover-neighbour highlighting, click-to-pin, click-to-isolate), **encoding** (size, color, community hulls), **rendering & scale** (semantic zoom, `foreignObject` labels, performance budget), and **off-main-thread** simulation with Web Workers.

**tutorials:** (index)

- [11-networks-d3js/11a-networkx-to-d3.html](#) — node-link JSON interchange; nodes/links contract; fixed circle and random-layout baselines
- [11-networks-d3js/11b-force-layouts.html](#) — `d3.forceSimulation`; link/charge/center/collide forces; tuning; pinning nodes
- [11-networks-d3js/11c-interactivity.html](#) — drag behaviour; hover-neighbour highlighting; click-to-pin; click-to-isolate
- [11-networks-d3js/11d-encoding-and-zoom.html](#) — visual encoding for graphs; community overlays via convex hulls
- [11-networks-d3js/11e-rendering-and-scale.html](#) — semantic zoom, `foreignObject` HTML labels, and rendering/performance techniques for scale
- [11-networks-d3js/11f-web-workers.html](#) — moving `d3.forceSimulation` off the main thread with a Web Worker; `Transferable` typed arrays; a UI responsiveness probe
- [11-networks-d3js/11g-capstone.html](#) — capstone: complete Game of Thrones viewer (force layout, colour, size, drag, hover, click-to-pin, semantic zoom, search, community filter, hulls)

**examples:** 18 self-contained HTML demos in [examples/](#) (linked from the index above), one per concept.

Legacy slide deck [dv11-D3-course-lesson-5.pdf](#) (third-party material) is kept under [slides/](#) as supplementary reference.

## Lecture 12: Advanced Network Visualization — Adjacency Matrix, Chord, Sankey, Edge Bundling, Dynamic Graphs, Beyond D3

Alternatives to the node-link diagram and where each one wins: **adjacency matrix** (ordering as the visualization), **chord diagram** (the matrix bent into a circle), **Sankey** (directed weighted flows; width = quantity), **edge bundling** (hierarchical and force-directed), **dynamic & streaming graphs** (keyed joins, gentle reheats, preferential-attachment growth, snapshot transitions), the **wider ecosystem** beyond D3 (Cytoscape.js, sigma.js + Graphology, Cosmograph, Gephi), and a **capstone** comparing four views of the same data under one synchronized hover.

**tutorials:** (index)

- [12-advanced-network-viz/12a-adjacency-matrix.html](#) —  $N \times N$  matrix from an edge list; row ordering (alphabetical / community / centrality / barycenter); row + column hover

- [12-advanced-network-viz/12b-chord-diagram.html](#) — `d3.chord` + `d3.ribbon` + `d3.arc`; community ordering on the rim; hover-to-isolate a node's flow
- [12-advanced-network-viz/12c-sankey.html](#) — the `d3-sankey` plugin; `nodeAlign`; width = quantity; linked Sankey + force views with synchronized hover
- [12-advanced-network-viz/12d-edge-bundling.html](#) — hierarchical edge bundling (`d3.cluster` + `curveBundle`) and force-directed bundling (`d3.ForceEdgeBundling`)
- [12-advanced-network-viz/12e-dynamic-graphs.html](#) — dynamic graphs: keyed joins, gentle reheats, preferential-attachment streaming (hub emergence), snapshot transitions
- [12-advanced-network-viz/12f-ecosystem.html](#) — beyond D3: Cytoscape.js, sigma.js + Graphology, Cosmograph, Gephi, vis-network, PyVis — and when to stay in D3
- [12-advanced-network-viz/12g-capstone-comparison.html](#) — capstone: force / matrix / chord / bundling on the same data, one synchronized hover, with a decision guide

**examples:** 9 self-contained HTML demos in [examples/](#) (linked from the index above), one per demo.

## References

Munzner, T. 2014. *Visualization Analysis and Design*. AK Peters Visualization Series. CRC Press. <https://books.google.it/books?id=dznSBQAAQBAJ>.